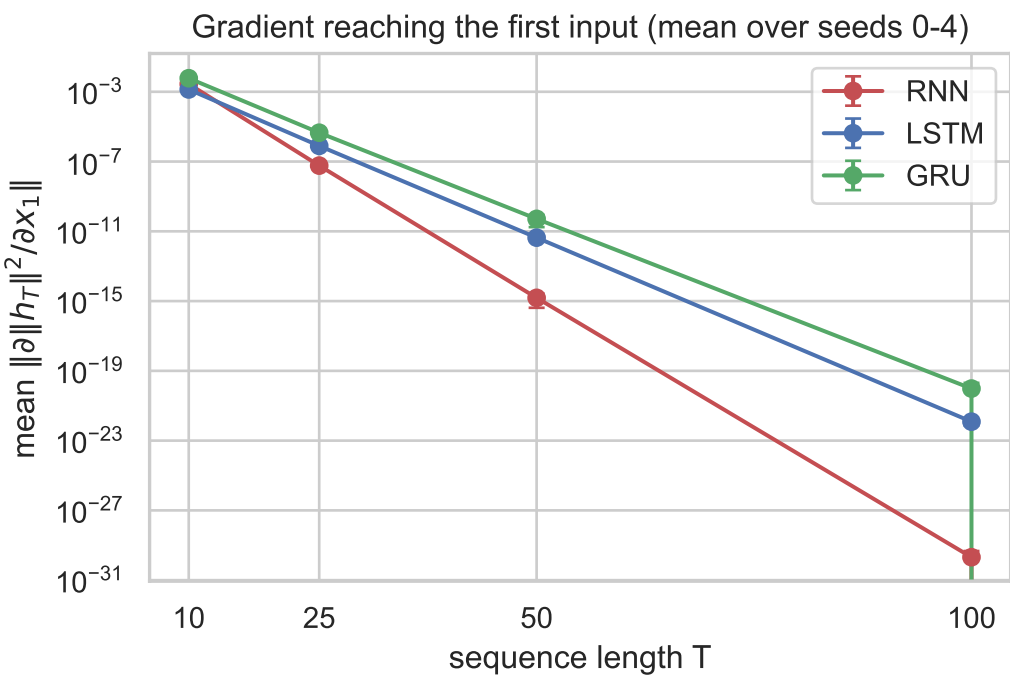
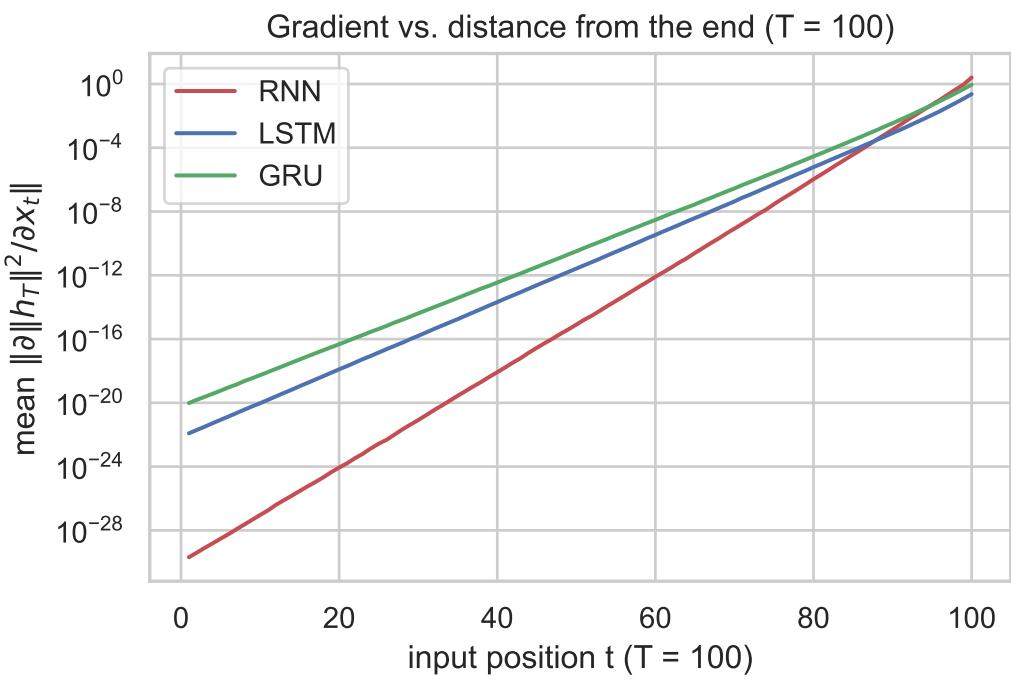

Gradient reaching the first input (mean over seeds 0-4)
RNN
LSTM
GRU
mean $\|\partial\|h_T\|^2/\partial x_1\|$
10^{-3}
10^{-7}
10^{-11}
10^{-15}
10^{-19}
10^{-23}
10^{-27}
10^{-31}
10
25
50
100
sequence length T


Gradient vs. distance from the end (T = 100)
RNN
LSTM
GRU
mean $\|\partial\|h_T\|^2/\partial x_t\|$
10^{0}
10^{-4}
10^{-8}
10^{-12}
10^{-16}
10^{-20}
10^{-24}
10^{-28}
0
20
40
60
80
100
input position t (T = 100)